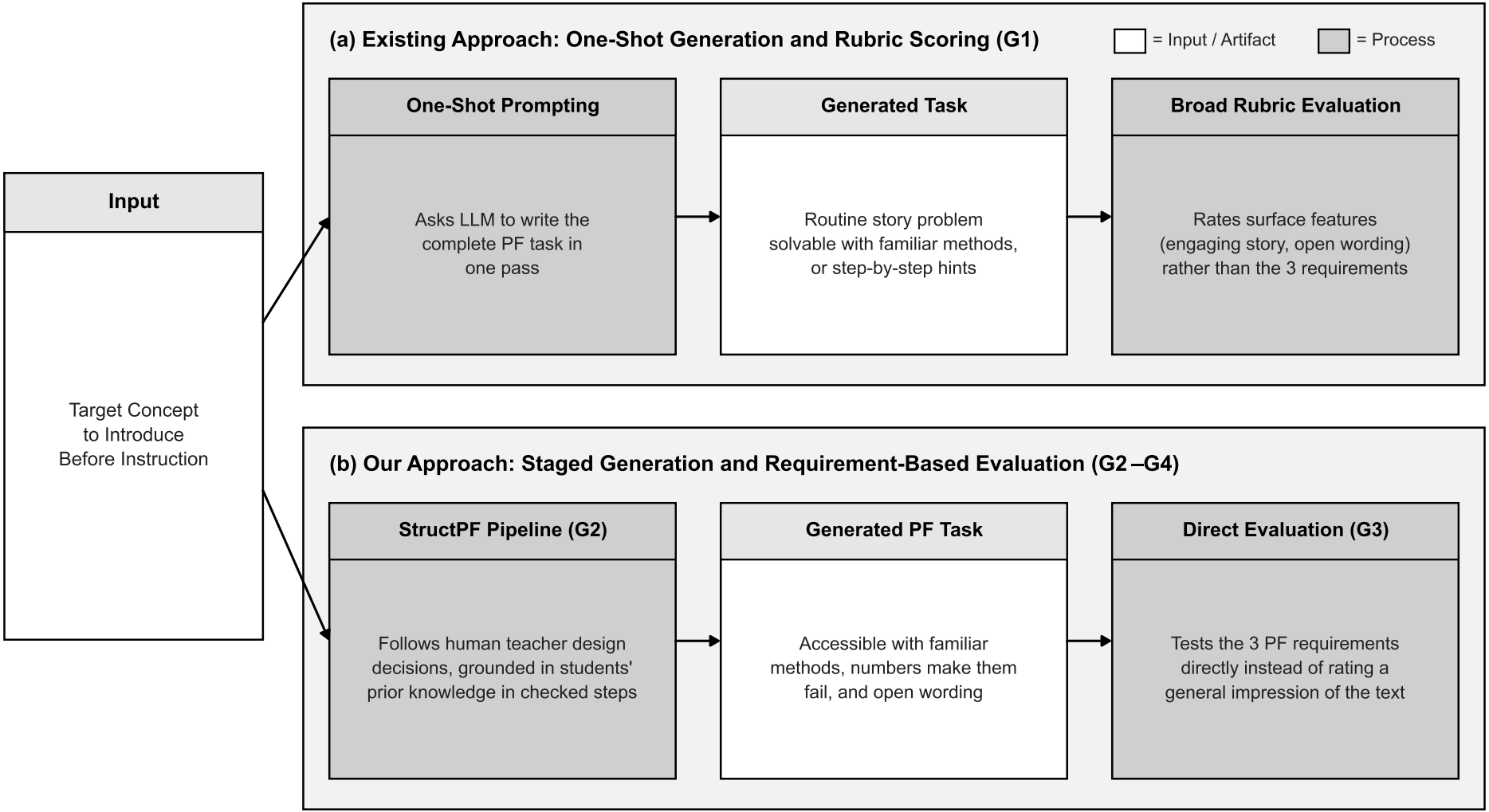
(a) Existing Approach: One-Shot Generation and Rubric Scoring (G1)
= Input / Artifact
= Process
Input
Target Concept to Introduce Before Instruction
One-Shot Prompting
Asks LLM to write the complete PF task in one pass
Generated Task
Routine story problem solvable with familiar methods, or step-by-step hints
Broad Rubric Evaluation
Rates surface features (engaging story, open wording) rather than the 3 requirements
(b) Our Approach: Staged Generation and Requirement-Based Evaluation (G2 –G4)
StructPF Pipeline (G2)
Follows human teacher design decisions, grounded in students' prior knowledge in checked steps
Generated PF Task
Accessible with familiar methods, numbers make them fail, and open wording
Direct Evaluation (G3)
Tests the 3 PF requirements directly instead of rating a general impression of the text